Quality Ladders & Schumpeterian Growth

Lecture 5 Cheatsheet — Macroeconomics I (BSE)

Exam Reference

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1 Notation and Definitions

Core Parameters and Variables

N = number of intermediate goods (fixed)
$k(s)$ = quality rung of good s
L = total labor (inelastic)
r = interest rate
σ = intertemporal elasticity of substitution
ξ = R&D cost parameter
w = wage
$V(k)$ = present value of innovation
$\gamma = g_Y$ = BGP growth rate

$q > 1$ = quality step size
$\alpha \in (0, 1)$ = intermediate input share
L_Y = labor in final good sector
ρ = household discount rate
$p(s)$ = innovation (arrival) rate
$Z(s)$ = R&D resources for good s
π = monopoly profits
Q = aggregate quality index

Key Derived Objects

$$\tilde{X}_t(s) := q^{k(s)} X_t(s) \quad \text{(quality-adjusted intermediate input)}$$

$$Q_t := \int_0^N q^{k(s)\frac{\alpha}{1-\alpha}} ds \quad \text{(aggregate quality index)}$$

$$\overline{\pi} := \frac{1-\alpha}{\alpha} \alpha^{\frac{2}{1-\alpha}} L \quad \text{(normalized per-sector profit)}$$

Key Assumption: Innovation Step Size

Critical Assumption

The model assumes $q \geq \frac{1}{\alpha}$. Under this assumption, the new innovator can always profitably undercut the old incumbent and become the *sole* monopolist. This justifies full creative destruction (no simultaneous coexistence of old and new producer).

Differences from Romer (Expanding Varieties)

Feature	**Romer (Varieties)**	**Aghion-Howitt (Quality)**
Growth mechanism	New goods (N grows)	Quality improvements (q rises)
N (varieties)	Grows over time	Fixed
Incumbent fate	Coexists with entrants	Destroyed by entrant
Business stealing	No	Yes
Creative destruction	No	Yes
Externality sign	Positive only	Positive + Negative
Net welfare bias	Under-innovation	Ambiguous

2 Model Setup: Final Good and Quality-Adjusted Intermediates

Final Good Production

The representative final-good firm solves:

$$\max_{L,\, X_t(s)} \; Y_t - w_t L - \int_0^N P_t(s)\, X_t(s)\, ds$$

subject to the technology:

$$Y_t = L^{1-\alpha} \int_0^N \left(q^{k(s)} X_t(s)\right)^{\alpha} ds \tag{Final Good}$$

where N is the (fixed) measure of intermediate goods and $q^{k(s)}$ embeds quality into productivity.

First-Order Conditions of Final Good Firm

Labor demand:

$$w_t = (1-\alpha)\, L^{-\alpha} \int_0^N \tilde{X}_t^{\alpha}(s)\, ds = (1-\alpha)\, \frac{Y_t}{L} \tag{Labor FOC}$$

Intermediate input demand (inverse demand for $X_t(s)$):

$$P_t(s) = \alpha\, L^{1-\alpha}\, q^{\alpha k(s)}\, X_t(s)^{\alpha-1} \tag{Demand}$$

Inverting the demand function:

$$\boxed{X_t(s) = L \left(\frac{\alpha\, q^{\alpha k(s)}}{P_t(s)} \right)^{\frac{1}{1-\alpha}}} \tag{Demand Inverted}$$

This is an isoelastic demand with price-elasticity $\dfrac{1}{1-\alpha}$.

Aggregation and Quality Index

Define the aggregate quality index:

$$Q_t := \int_0^N q^{k(s)\frac{\alpha}{1-\alpha}}\, ds$$

Then aggregate output simplifies to:

$$\boxed{Y_t = \alpha^{\frac{2\alpha}{1-\alpha}}\, L\, Q_t} \tag{Aggregate Output}$$

Implication: $Y_t \propto Q_t$, so the growth rate of output equals the growth rate of the quality index on the BGP.

Resource Constraint

$$Y_t = C_t + X_t + Z_t$$

where $X_t = \int_0^N X_t(s)\, ds$ (total intermediates) and $Z_t = \int_0^N Z_t(k(s))\, ds$ (total R&D).

3 Intermediate Monopolist: Pricing and Profits

Setup

The successful innovator gains a quality advantage over all previous producers:

- **Old incumbent**: produces quality q^{k-1}, marginal cost $= 1$
- **New innovator**: produces quality q^k, marginal cost $= 1$
- Quality improvement ratio: q (the innovation step)

The new innovator solves:

$$\max_{X_t(s)} \pi_t(s) = [P_t(s) - 1]\, X_t(s) \quad \text{subject to demand } X_t(s) = L\left(\frac{\alpha q^{\alpha k(s)}}{P_t(s)}\right)^{\frac{1}{1-\alpha}}$$

Unconstrained Monopoly Price

Step 1. Substitute demand into profits:

$$\pi = (P-1) \cdot L\left(\frac{\alpha q^{\alpha k}}{P}\right)^{\frac{1}{1-\alpha}}$$

Step 2. Take FOC with respect to P:

$$\frac{d\pi}{dP} = 0 \implies \frac{1}{P} \cdot \frac{\alpha}{1-\alpha} = \frac{1}{P-1} \cdot \frac{1}{1-\alpha}$$

Step 3. Simplify to get the standard markup rule (marginal cost $= 1$, demand elasticity $= 1/(1-\alpha)$):

$$P^* = \frac{1}{\alpha}$$

This is the standard markup pricing formula: $P = \text{MC} \times \frac{1/(1-\alpha)}{1/(1-\alpha)-1} = \frac{1}{\alpha}$.

Limit Pricing Argument

The new innovator can threaten to undercut the old producer. In quality-adjusted terms, the old producer's effective price is $\frac{P^{\text{old}}}{q^{k-1}}$ whereas the new producer's effective price is $\frac{P^{\text{new}}}{q^k}$. The new innovator captures the entire market if:

$$\frac{P^{\text{new}}}{q^k} < \frac{P^{\text{old}}}{q^{k-1}} \iff P^{\text{new}} < q \cdot P^{\text{old}}$$

The old incumbent's lowest possible price is $P^{\text{old}} = 1$ (marginal cost). So the limit price constraint is:

$$P^{\text{new}} \leq q \quad \text{(limit price)}$$

The Two Cases: $q \geq 1/\alpha$ vs. $q < 1/\alpha$

- **Case 1: $q \geq 1/\alpha$ (assumed in main model)**
  Unconstrained monopoly price $P^* = 1/\alpha \leq q$.
  Limit pricing does NOT bind. Innovator charges $P = 1/\alpha$ and is unchallenged.

- **Case 2: $q < 1/\alpha$ (limit pricing binds)**
  If innovator charged $1/\alpha > q$, the old incumbent could undercut.
  Innovator must charge $P = q$ (Bertrand competition with old producer).
  Old and new producer may coexist.

Monopoly Profits

Step 1. With price $P = 1/\alpha$ and MC $= 1$, the markup is:

$$P - 1 = \frac{1}{\alpha} - 1 = \frac{1-\alpha}{\alpha}$$

Step 2. Quantity demanded at $P = 1/\alpha$:

$$X(k(s)) = L\,\alpha^{\frac{2}{1-\alpha}}\,q^{\frac{\alpha k(s)}{1-\alpha}}$$

Step 3. Profits per sector s at rung $k(s)$:

$$\pi(k(s)) = \frac{1-\alpha}{\alpha}\,\alpha^{\frac{2}{1-\alpha}}\,L\,q^{\frac{\alpha k(s)}{1-\alpha}} = \overline{\pi}\,q^{\frac{\alpha k(s)}{1-\alpha}}$$

where $\overline{\pi} := \frac{1-\alpha}{\alpha}\,\alpha^{\frac{2}{1-\alpha}}\,L$.

Step 4. In aggregate (using $Y_t = \alpha^{\frac{2\alpha}{1-\alpha}} L Q_t$):

$$\pi_{\text{total}} = (1-\alpha)\,Y_t/N \quad \text{(per-sector normalized)}$$

$$\boxed{P = \frac{1}{\alpha}, \quad \pi(k(s)) = \overline{\pi}\,q^{\frac{\alpha k(s)}{1-\alpha}}, \quad \overline{\pi} = \frac{1-\alpha}{\alpha}\,\alpha^{\frac{2}{1-\alpha}}\,L} \qquad \text{(Monopoly)}$$

4 Innovation: R&D, Value of Innovation, and Bellman Equation

Innovation Technology

A potential entrant invests Z units of output to achieve an innovation at Poisson rate p. The innovation probability is specified as:

$$p(k(s)) = \frac{Z(k(s))}{\xi\, q^{\frac{\alpha(k(s)+1)}{1-\alpha}}}$$

This means higher quality rungs require more resources per unit of innovation probability (innovations are harder at the frontier).

The innovator's problem is:

$$\max_{Z\geq 0}\; p(k(s))\, V(k(s)+1) - Z = \frac{Z}{\xi\, q^{\frac{\alpha(k(s)+1)}{1-\alpha}}}\, V(k(s)+1) - Z$$

Value of Innovation and Bellman Equation

Once the innovator holds the patent at rung k, they earn flow profits $\pi(k)$ until replaced by the next innovator arriving at Poisson rate p (creative destruction). With a constant interest rate r:

Step 1. In continuous time, the value of the patent satisfies the asset pricing (HJB) equation:

$$r\, V(k) = \pi(k) - p\, V(k) + \dot{V}$$

The incumbent earns $\pi(k)$ per unit time, loses $V(k)$ when an innovation arrives (rate p), and $\dot{V}$ captures capital gains.

Step 2. On the BGP, V grows at the same rate as π and Y, i.e., $\dot{V}/V = \gamma$. Rearranging:

$$(r + p - \gamma)\, V(k) = \pi(k) \implies V(k) = \frac{\pi(k)}{r + p - \gamma}$$

Step 3. Simplification (constant π, $\dot{V} = 0$, constant rate r):

$$V(k) = \frac{\pi(k)}{r + p}$$

Step 4. Using $\pi(k(s)) = \overline{\pi}\, q^{\frac{\alpha k}{1-\alpha}}$:

$$V(k(s)) = \frac{\overline{\pi}\, q^{\frac{\alpha k(s)}{1-\alpha}}}{r + p}$$

$$\boxed{r\, V = \pi - p\, V + \dot{V}} \quad \text{(Bellman / no-arbitrage)} \qquad \text{(HJB)}$$

$$\boxed{V(k) = \frac{\pi(k)}{r + p}} \quad \text{(BGP with } \dot{V} = 0\text{)} \qquad \text{(Value)}$$

Free Entry Condition into R&D

With free entry, the expected return to R&D equals its cost. The FOC of the innovator's problem gives:

$$\frac{V(k(s)+1)}{\xi\, q^{\frac{\alpha(k(s)+1)}{1-\alpha}}} = 1 \qquad \text{(Free Entry)}$$

Since $V(k+1) = \overline{\pi}\, q^{\frac{\alpha(k+1)}{1-\alpha}}/(r+p)$, this simplifies to:

$$\frac{\overline{\pi}}{\xi(r+p)} = 1 \implies \boxed{r = \frac{\overline{\pi}}{\xi} - p} \qquad \text{(Free Entry Simplified)}$$

This is a key relationship: the interest rate equals the ratio of normalized profits to R&D costs, minus the innovation rate.

Equilibrium Innovation Rate

The optimal R&D investment per sector is:

$$Z^*(k(s)) = \xi\, q^{\frac{\alpha(k(s)+1)}{1-\alpha}}\, p = q^{\frac{\alpha(k(s)+1)}{1-\alpha}}(\overline{\pi} - r\xi)$$

Key insight: The innovation arrival rate $p = \overline{\pi}/\xi - r$ is *identical across all sectors*, regardless of the current quality rung. Higher-rung sectors simply require more resources per unit of p.

5 Free Entry, Euler Equation, and BGP Growth Rate

Quality Growth Dynamics

Each sector independently improves quality by a factor q at Poisson rate p. Using the law of large numbers across sectors:

$$\frac{\dot{Q}}{Q} = p\left[q^{\frac{\alpha}{1-\alpha}} - 1\right] \tag{Quality Growth}$$

Since $Y_t \propto Q_t$, the BGP growth rate of output is:

$$\boxed{\gamma = \frac{\dot{Y}}{Y} = \frac{\dot{Q}}{Q} = p\left[q^{\frac{\alpha}{1-\alpha}} - 1\right]} \tag{BGP Output Growth}$$

Household Euler Equation

Households maximize $\int_0^\infty \frac{c^{1-\sigma}-1}{1-\sigma} e^{-\rho t} dt$, yielding:

$$\frac{\dot{c}}{c} = \frac{1}{\sigma}(r - \rho) \tag{Euler}$$

On the BGP: $\dot{c}/c = \gamma$, so:

$$r = \rho + \sigma\gamma$$

Solving for the BGP

Step 1. From free entry: $r = \overline{\pi}/\xi - p$.
Step 2. From Euler ($\sigma = 1$): $r = \rho + \gamma$.
Step 3. Combine:

$$\rho + \gamma = \frac{\overline{\pi}}{\xi} - p$$

Step 4. From quality growth: $\gamma = p(q^{\frac{\alpha}{1-\alpha}} - 1)$. Substitute:

$$\rho + p\left(q^{\frac{\alpha}{1-\alpha}} - 1\right) = \frac{\overline{\pi}}{\xi} - p \implies p\,q^{\frac{\alpha}{1-\alpha}} = \frac{\overline{\pi}}{\xi} - \rho$$

$$\boxed{p = \left(\frac{\overline{\pi}}{\xi} - \rho\right) q^{-\frac{\alpha}{1-\alpha}}}$$

Step 5. Substitute back for growth rate:

$$\gamma = p\left(q^{\frac{\alpha}{1-\alpha}} - 1\right) = \left(\frac{\overline{\pi}}{\xi} - \rho\right)\left(1 - q^{-\frac{\alpha}{1-\alpha}}\right)$$

$$\boxed{p^* = \left(\frac{\overline{\pi}}{\xi} - \rho\right) q^{-\frac{\alpha}{1-\alpha}}} \qquad \text{(Eqm Innovation Rate)}$$

$$\boxed{\gamma^* = \left(\frac{\overline{\pi}}{\xi} - \rho\right)\left(1 - q^{-\frac{\alpha}{1-\alpha}}\right)} \qquad \text{(BGP Growth)}$$

$$\boxed{r^* = \frac{\overline{\pi}}{\xi}\left(1 - q^{-\frac{\alpha}{1-\alpha}}\right) + \rho\, q^{-\frac{\alpha}{1-\alpha}}} \qquad \text{(Eqm Interest Rate)}$$

BGP Existence Condition

For $p > 0$ and $\gamma > 0$, we need:

$$\frac{\overline{\pi}}{\xi} > \rho \quad \text{(profits must be large relative to discounting)}$$

6 Market Equilibrium vs Social Planner

Planner's Problem

The social planner maximizes household welfare subject to resource and technology constraints:

$$\max\ L \int_0^\infty \frac{c_t^{1-\sigma} - 1}{1-\sigma} e^{-\rho t} dt$$

subject to $Y_t = X_t + Z_t + Lc_t$ and $\dot{Q}_t = \dfrac{Z_t(1 - q^{-\frac{\alpha}{1-\alpha}})}{\xi}$.

Planner's Optimal Allocations

The planner sets price equal to marginal cost ($P^* = 1$ vs market $P = 1/\alpha$), so:

$$X_t^*(s) = L\,\alpha^{\frac{1}{1-\alpha}}\,q^{\frac{\alpha k(s)}{1-\alpha}} > X_t(s) = L\,\alpha^{\frac{2}{1-\alpha}}\,q^{\frac{\alpha k(s)}{1-\alpha}}$$

The planner uses *more* intermediates than the market (no monopoly markup distortion). Correspondingly:

$$Y_t^* = L\,\alpha^{\frac{\alpha}{1-\alpha}} Q_t > Y_t = L\,\alpha^{\frac{2\alpha}{1-\alpha}} Q_t \quad \text{(at given } Q_t\text{)}$$

The planner's implicit profit from innovation is:

$$\pi^* = \frac{1-\alpha}{\alpha}\,\alpha^{\frac{1}{1-\alpha}} L > \overline{\pi} = \frac{1-\alpha}{\alpha}\,\alpha^{\frac{2}{1-\alpha}} L \qquad \text{(Planner Profit)}$$

Since $\alpha^{\frac{1}{1-\alpha}} > \alpha^{\frac{2}{1-\alpha}}$ for $\alpha \in (0,1)$.

Planner's Growth Rate

Using the Hamiltonian on the planner's problem yields the planner's Euler equation with implicit interest rate:

$$r^* = \frac{\pi^*}{\xi}\left(1 - q^{-\frac{\alpha}{1-\alpha}}\right)$$

And the planner's growth rate:

$$\gamma^* = \frac{1}{\sigma}\left[\frac{\pi^*}{\xi}\left(1 - q^{-\frac{\alpha}{1-\alpha}}\right) - \rho\right] \qquad \text{(Planner Growth)}$$

Comparing Market and Planner Rates

Market interest rate:

$$r = \frac{\overline{\pi}}{\xi}\left(1 - q^{-\frac{\alpha}{1-\alpha}}\right) + \rho\, q^{-\frac{\alpha}{1-\alpha}}$$

Externality 1 — Business Stealing (negative for welfare):
An entrant earns the *full* value $V(k+1)$, ignoring that it destroys the incumbent's profits $V(k)$. In the market, the innovator captures the entire monopoly rent without internalizing the loss imposed on the previous incumbent. This creates an incentive for *excessive* R&D. The extra term $\rho\, q^{-\frac{\alpha}{1-\alpha}} > 0$ in r captures this: market interest is too high, so market innovations p are too low given households' discounting — but entrants innovate *too much* relative to the social value of destroying incumbents.
Externality 2 — Appropriability / Knowledge Spillover (positive for welfare):
Innovators capture only monopoly profits $\overline{\pi}$, but the social value of innovation includes consumer surplus (the difference between WTP and price). The planner values innovation at $\pi^* > \overline{\pi}$ because it sets $P = MC = 1$ rather than $P = 1/\alpha$. This creates an incentive for *insufficient* R&D in the market.
Net Effect:

$$\gamma^* \gtrless \gamma \quad \text{(ambiguous)}$$

Business stealing $\Rightarrow$ tendency toward over-innovation in market.
Appropriability externality $\Rightarrow$ tendency toward under-innovation in market.
Net direction depends on parameters (q, α, ρ).

Contrast with Romer: In expanding varieties models, only the appropriability (positive) externality exists — no business stealing since old varieties coexist. Result: unambiguous under-innovation.

Policy Implications

- A tax τ on R&D raises the cost of entry and reduces p and γ.
- Crucially: *current incumbents favor taxing R&D* because it protects their market position.
- This creates a political economy problem: those with most influence (current monopolists) have the strongest incentive to suppress innovation.
- If the economy has too much growth, taxing R&D may be Pareto-improving. If too little, subsidizing may be.

7 Competition and Innovation: Escape Competition Effect

Step-by-Step Innovation with Duopolists

The Aghion-Howitt (AH) framework can be extended to allow *neck-and-neck* competition between two incumbent firms:

- Each product line j is supplied by a duopoly $\{\alpha, \beta\}$ plus a competitive fringe.
- Firms' productivities: $A_i = \gamma^{k_i}$, with competitors at most one step apart.
- Two types of sectors:
  - **Neck-and-neck** ($\Delta k = 0$): both firms at same rung, fringe one step below.
  - **Unlevel** ($\Delta k = 1$): leader one step ahead of follower, fringe two steps below.

Bertrand Competition and Payoffs

Under Bertrand competition:

- **Unlevel sector**: Leader prices at follower's marginal cost $\Rightarrow$ markup $= \gamma$.

$$\pi_1 = \frac{\gamma - 1}{\gamma}, \quad \pi_{-1} = 0$$

- **Neck-and-neck sector**: Each firm gets

$$\pi_0 = (1 - \Delta)\pi_1$$

  where $\Delta \in [0, 1]$ is the competition intensity ($\Delta \to 1$: full competition, $\pi_0 \to 0$).

Innovation Incentives

Unlevel sector (laggard innovates at rate n_{-1}):

$$\max_{n_{-1}} (n_{-1} + h)\pi_0 - \frac{\psi n_{-1}^2}{2} \implies n_{-1} = (1 - \Delta)\frac{\pi_1}{\psi}$$

More competition ($\Delta \uparrow$) $\Rightarrow n_{-1} \downarrow$ (Schumpeterian effect).

Neck-and-neck sector (one firm innovates at rate n_0):

$$\max_{n_0} n_0\pi_1 + (1 - n_0)\pi_0 - \frac{\psi n_0^2}{2} \implies n_0 = \frac{\Delta\,\pi_1}{\psi}$$

More competition ($\Delta \uparrow$) $\Rightarrow n_0 \uparrow$ (**escape competition effect**).

Inverted-U Relationship

In steady state, let μ_1 = fraction of unlevel sectors:

$$\mu_1 = \frac{n_0}{n_0 + n_{-1} + h}$$

The total innovation flow is:

$$I = \frac{2\pi_1}{\psi\left(1 + \frac{\psi h}{\pi_1}\right)} \Delta\left(1 - \Delta + \frac{\psi h}{\pi_1}\right)$$

Derivative with respect to Δ:

$$\frac{\partial I}{\partial \Delta} \propto 1 - 2\Delta + \frac{\psi h}{\pi_1}$$

Low competition (Δ small): Most sectors neck-and-neck $\Rightarrow$ escape competition effect dominates $\Rightarrow \partial I/\partial \Delta > 0$.
High competition (Δ near 1): Most sectors unlevel $\Rightarrow$ Schumpeterian effect dominates $\Rightarrow \partial I/\partial \Delta < 0$.
Result: Non-monotonic (inverted-U) relationship between competition Δ and innovation I.

Key finding (Aghion et al., 2005): Competition and innovation have an *inverted-U* relationship:

- At low competition: more competition $\Rightarrow$ more innovation (escape competition dominates)
- At high competition: more competition $\Rightarrow$ less innovation (Schumpeterian effect dominates)

Empirical evidence: UK firm-level data 1973–1994 (citation-weighted patents vs Lerner index).

8 Comparison: Expanding Varieties vs Quality Ladders

Feature	Expanding Varieties (Romer)	Quality Ladders (Aghion-Howitt)
Growth mechanism	New goods created; N grows	Quality improvements; N fixed
Horizontal / vertical	Horizontal differentiation	Vertical differentiation
Creative destruction	No: new goods coexist with old	Yes: new innovator displaces incumbent
Business stealing	No (new variety adds to market)	Yes (entrant destroys incumbent's value)
Number of varieties	N grows over time	N constant
Incumbent fate	Keeps market share, earns forever	Destroyed by next innovator
Welfare bias	Under-innovation (appropriability only)	Ambiguous (two opposing externalities)
Policy prescription	Subsidize R&D	Depends on which externality dominates
Political economy	Incumbents neutral on innovation	Incumbents actively resist innovation
Growth rate depends on	Size of N-expansion technology	Quality step q, R&D productivity $1/\xi$

Mathematical Correspondence

In both models, the BGP growth rate has a similar structure:

$$\gamma_{\text{Romer}} = \frac{\lambda L - \rho\eta}{\sigma\eta + 1} \quad \text{(from free entry + Euler)}$$

$$\gamma_{\text{AH}} = \left(\frac{\overline{\pi}}{\xi} - \rho\right)\left(1 - q^{-\frac{\alpha}{1-\alpha}}\right) \quad \text{(log utility)}$$

Both are *scale-effect models*: growth depends on total labor L. Both require R&D profits to cover discounting. Both pin down growth through the free-entry + Euler system.

Key Difference: Conflict of Interest

The Political Economy Difference

In Expanding Varieties: no conflict between incumbents and entrants. Everyone benefits from more innovation.

In Quality Ladders: innovation *destroys* incumbent profits. Current monopolists have an incentive to lobby against R&D policies (e.g., support R&D taxes), even if such policies reduce aggregate welfare. This creates a conflict of interest between incumbents (losers from innovation) and consumers/entrants (gainers from innovation).

9 Problem-Solving Templates

(a) Derive Monopoly Price and Profits

1. Write down the final good firm's demand for intermediate s:

$$P_t(s) = \alpha L^{1-\alpha} q^{\alpha k(s)} X_t(s)^{\alpha-1}$$

2. The monopolist sets $\max_X (P(X) - 1)X$. Use the isoelastic demand to get markup $P = \text{MC}/\alpha = 1/\alpha$.
3. Compute equilibrium quantity: $X^* = L\alpha^{\frac{2}{1-\alpha}} q^{\frac{\alpha k}{1-\alpha}}$.
4. Compute profits: $\pi = (P-1)X = \frac{1-\alpha}{\alpha} X = \overline{\pi} q^{\frac{\alpha k}{1-\alpha}}$.
5. **Check** whether $q \geq 1/\alpha$: if yes, limit pricing does not bind; if no, use $P = q$ instead.

(b) Set Up and Solve the Bellman Equation for V

1. Write the no-arbitrage HJB: $r\,V = \pi - p\,V + \dot{V}$.
2. On the BGP, V grows at rate γ, so $\dot{V} = \gamma V$.
3. Rearrange: $(r + p - \gamma)V = \pi \implies V = \pi/(r + p - \gamma)$.
4. Simplified version (often used): $V = \pi/(r + p)$ when $\dot{V} = 0$.
5. Apply free entry: $V(k+1)/[\xi q^{\frac{\alpha(k+1)}{1-\alpha}}] = 1$, which gives $\overline{\pi}/[\xi(r + p)] = 1$.

(c) Apply Free Entry + Euler for BGP Growth

1. **Free entry:** $r = \overline{\pi}/\xi - p$ (from innovator's FOC).
2. **Quality growth:** $\gamma = p(q^{\frac{\alpha}{1-\alpha}} - 1)$.
3. **Euler equation:** $r = \rho + \sigma\gamma$ (households' optimality).
4. Substitute Euler into free entry: $\rho + \sigma\gamma = \overline{\pi}/\xi - p$.
5. Substitute quality growth for γ: solve for p and γ simultaneously.
6. For $\sigma = 1$: $p = (\overline{\pi}/\xi - \rho)\, q^{-\frac{\alpha}{1-\alpha}}$ and $\gamma = (\overline{\pi}/\xi - \rho)(1 - q^{-\frac{\alpha}{1-\alpha}})$.

(d) Identify and Sign Externalities

1. Solve the planner's problem: Hamiltonian with Q as state, Z (or C) as control.
2. Planner sets $P = MC = 1$ (no markup), yielding $\pi^* > \overline{\pi}$.
3. Compare r^* (planner) vs r (market):
   - Term $(\pi^*/\xi - \overline{\pi}/\xi)$ reflects appropriability externality (market under-values innovation).
   - Term $\rho q^{-\frac{\alpha}{1-\alpha}} > 0$ reflects business-stealing (market over-values innovation).
4. Sign the net externality: $\gamma^* \gtrless \gamma$ depends on whether appropriability or business-stealing dominates.
5. State policy direction: subsidize R&D if $\gamma < \gamma^*$; tax if $\gamma > \gamma^*$.

(e) Analyze Competition and Innovation (Escape Competition)

1. Set up payoffs for neck-and-neck ($\pi_0 = (1-\Delta)\pi_1$) and unlevel (π_1, 0) sectors.
2. Solve innovator's optimization in each sector type: get $n_0 = \Delta\pi_1/\psi$ and $n_{-1} = (1-\Delta)\pi_1/\psi$.
3. Compute steady-state sector composition μ_1 and aggregate innovation flow I.
4. Differentiate I w.r.t. Δ and sign: $\partial I/\partial\Delta \gtrless 0$ depending on whether $\Delta \lessgtr (1+\psi h/\pi_1)/2$.
5. Conclude: inverted-U relationship. Identify which regime the economy is in.

10 Common Pitfalls

1. **Confusing the two q cases for limit pricing**
   - If $q \geq 1/\alpha$: monopolist charges $P = 1/\alpha$; limit pricing not binding.
   - If $q < 1/\alpha$: monopolist charges $P = q$; old and new may coexist.
   - Always check which case applies before deriving profits.
2. **Forgetting that N is fixed**
   - Unlike Romer where N grows, in AH the number of sectors N is constant.
   - Growth comes *entirely* from quality improvements, not variety expansion.
3. **Misidentifying the direction of externalities**
   - Business stealing: entrant ignores *destroyed* incumbent value $\Rightarrow$ over-innovation tendency.
   - Appropriability: innovator cannot capture full consumer surplus $\Rightarrow$ under-innovation tendency.
   - Many students reverse these; remember: "stealing" refers to the externality *on the incumbent*.
4. **Claiming unambiguous over- or under-innovation**
   - In AH, the result is *ambiguous*: either $\gamma > \gamma^*$ or $\gamma < \gamma^*$ is possible.
   - Only in Romer (expanding varieties) is the direction unambiguous (under-innovation).
5. **Errors in the Bellman equation**
   - The HJB on the BGP: $r\,V = \pi - p\,V + \dot{V}$. Do NOT omit the $\dot{V}$ term unless you argue it is zero.
   - On the BGP $\dot{V}/V = \gamma$, so $V = \pi/(r + p - \gamma)$, not $\pi/(r + p)$.
   - The simplified $V = \pi/(r + p)$ is valid when $\dot{V} = 0$ (constant V), which holds if π is constant and r is constant.
6. **Confusing escape competition with Schumpeterian effects**
   - Escape competition applies to *neck-and-neck* sectors: competition increases innovation.
   - Schumpeterian effect applies to *unlevel* sectors: competition reduces innovation.
   - Do not say "competition always reduces innovation" — it depends on sector type.
7. **Mixing up p (innovation rate) and γ (growth rate)**
   - p is the Poisson arrival rate of innovations (a frequency).
   - γ is the growth rate of output/quality: $\gamma = p(q^{\frac{\alpha}{1-\alpha}} - 1)$.
   - They are proportional but not equal.
8. **Scale effects**
   - Both Romer and AH exhibit scale effects: γ depends on total L.
   - This is a known shortcoming; Jones (1995) semi-endogenous growth removes it.

11 Quick Reference Formulas

Production and Demand

$$Y_t = L^{1-\alpha} \int_0^N (q^{k(s)} X_t(s))^\alpha ds \qquad \text{(Final Good)}$$

$$P_t(s) = \alpha L^{1-\alpha} q^{\alpha k(s)} X_t(s)^{\alpha-1} \qquad \text{(Intermediate Demand)}$$

$$Y_t = \alpha^{\frac{2\alpha}{1-\alpha}} L\, Q_t, \quad Q_t = \int_0^N q^{k(s)\frac{\alpha}{1-\alpha}} ds \qquad \text{(Aggregate Output)}$$

Monopoly Pricing and Profits

$$P = \frac{1}{\alpha} \quad (\text{if } q \geq 1/\alpha) \qquad \text{(Monopoly Price)}$$

$$\pi(k(s)) = \overline{\pi}\, q^{\frac{\alpha k(s)}{1-\alpha}}, \quad \overline{\pi} = \frac{1-\alpha}{\alpha} \alpha^{\frac{2}{1-\alpha}} L \qquad \text{(Monopoly Profit)}$$

Innovation and Value

$$r\,V = \pi - p\,V + \dot{V} \qquad \text{(Bellman / HJB)}$$

$$V(k) = \frac{\pi(k)}{r+p} \quad (\dot{V} = 0 \text{ case}) \qquad \text{(Value)}$$

$$\frac{\overline{\pi}}{\xi(r+p)} = 1 \quad (\text{free entry}) \implies r = \frac{\overline{\pi}}{\xi} - p \qquad \text{(Free Entry)}$$

BGP (log utility)

$$\frac{\dot{Q}}{Q} = p\Big(q^{\frac{\alpha}{1-\alpha}} - 1\Big) = \gamma \qquad \text{(Quality Growth)}$$

$$p^* = \left(\frac{\overline{\pi}}{\xi} - \rho\right) q^{-\frac{\alpha}{1-\alpha}} \qquad \text{(Eqm } p\text{)}$$

$$\gamma^* = \left(\frac{\overline{\pi}}{\xi} - \rho\right)\Big(1 - q^{-\frac{\alpha}{1-\alpha}}\Big) \qquad \text{(BGP Growth)}$$

$$r^* = \frac{\overline{\pi}}{\xi}\Big(1 - q^{-\frac{\alpha}{1-\alpha}}\Big) + \rho\, q^{-\frac{\alpha}{1-\alpha}} \qquad \text{(Eqm } r\text{)}$$

Planner vs Market

$$\pi^* = \frac{1-\alpha}{\alpha} \alpha^{\frac{1}{1-\alpha}} L > \overline{\pi} \qquad \text{(Planner Profit)}$$

$$r^*_{\text{planner}} = \frac{\pi^*}{\xi}\Big(1 - q^{-\frac{\alpha}{1-\alpha}}\Big) \qquad \text{(Planner Rate)}$$

$$\gamma^*_{\text{planner}} \gtrless \gamma^*_{\text{market}} \quad (\text{ambiguous}) \qquad \text{(Welfare)}$$

Escape Competition (Duopoly Extension)

$$n_0 = \frac{\Delta\,\pi_1}{\psi}, \quad n_{-1} = \frac{(1-\Delta)\pi_1}{\psi} \tag{Innovation Rates}$$

$$I = \frac{2\pi_1}{\psi\left(1+\frac{\psi h}{\pi_1}\right)}\,\Delta\left(1-\Delta+\frac{\psi h}{\pi_1}\right) \tag{Aggregate Innovation}$$

$$\frac{\partial I}{\partial \Delta} \propto 1 - 2\Delta + \frac{\psi h}{\pi_1} \gtrless 0 \tag{Inverted-U}$$

Comparison Table: Key Formulas

Object	**Romer (EV)**	**AH (QL)**
Growth driver	$\dot{N}/N$	$p(q^{\frac{\alpha}{1-\alpha}}-1)$
Free entry pins	$V_t = \eta w_t$	$\overline{\pi}/[\xi(r+p)] = 1$
Welfare bias	Under-innovation	Ambiguous
π vs π^*	$\pi < \pi^*$	$\overline{\pi} < \pi^*$
Business stealing	None	Yes